WORKED SOLUTION

STEP I 1995, Question 13

Probability and Statistics • Poisson Thinning

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q13

The key idea

Part (i) is the *thinning* property of the Poisson distribution: condition on the true number of cancerous cells, and the marked cells are then binomial. Summing over the unknown true number collapses neatly into an exponential series. For part (ii), the event "the second mark is on slide k" is decided by how many marks there are on the first $k-1$ slides altogether (zero or one), and part (i) together with the additivity of independent Poisson variables tells us exactly how that total is distributed.

Solution

Throughout, write $q = 1 - p$ and $\lambda = \mu p$. We take the numbers of cancerous cells on different slides to be independent (the natural reading of "random"), and the scientist's detections to be independent of each other and of the cell counts.

Part (i)

On one slide let N be the number of cancerous cells and M the number she marks, so $N \sim \mathrm{Po}(\mu)$. Given $N = n$, each of the n cells is spotted independently with probability p, so $M \mid N = n \sim \mathrm{B}(n, p)$. For $m = 0, 1, 2, \ldots$, by the law of total probability (only $n \geq m$ contributes),

$$\begin{aligned} \mathrm{P}(M = m) &= \sum_{n=m}^{\infty} \mathrm{e}^{-\mu} \frac{\mu^n}{n!} \binom{n}{m} p^m q^{n-m} = \mathrm{e}^{-\mu} \frac{(\mu p)^m}{m!} \sum_{n=m}^{\infty} \frac{(\mu q)^{n-m}}{(n-m)!} \\ &= \mathrm{e}^{-\mu} \frac{(\mu p)^m}{m!} \mathrm{e}^{\mu q} = \mathrm{e}^{-\mu p} \frac{(\mu p)^m}{m!}, \end{aligned}$$

using $\binom{n}{m}/n! = 1/(m!\,(n-m)!)$, then the exponential series with $j = n - m$, and finally $-\mu + \mu q = -\mu p$. So $M \sim \mathrm{Po}(\mu p)$, as required.

Part (ii)

Let M_i be the number of marked cells on slide i. By part (i) and independence between slides, $M_1, M_2, \ldots$ are independent $\mathrm{Po}(\lambda)$ variables. Let $S = M_1 + \cdots + M_{k-1}$ be the number marked on the first $k-1$ slides; a sum of independent Poisson variables is Poisson with the means added, so $S \sim \mathrm{Po}((k-1)\lambda)$, and S is independent of M_k.

> **Ask yourself** Exactly what has to happen for the second mark to land on slide k?

Before slide k she must have made at most one mark, and by the end of slide k at least two. So there are two disjoint cases:



either $S = 0$ and $M_k \geq 2$, or $S = 1$ and $M_k \geq 1$

(For $k = 1$ there are no earlier slides, $S = 0$ automatically, and the formulas below still hold.) Using the independence of S and M_k:

$$\mathrm{P}(S = 0,\ M_k \geq 2) = \mathrm{e}^{-(k-1)\lambda}\left(1 - \mathrm{e}^{-\lambda} - \lambda \mathrm{e}^{-\lambda}\right),$$
$$\mathrm{P}(S = 1,\ M_k \geq 1) = (k-1)\lambda\, \mathrm{e}^{-(k-1)\lambda}\left(1 - \mathrm{e}^{-\lambda}\right).$$

Adding,

$$Q = \mathrm{e}^{-(k-1)\lambda}\left\{\left(1 + (k-1)\lambda\right)\left(1 - \mathrm{e}^{-\lambda}\right) - \lambda \mathrm{e}^{-\lambda}\right\}.$$

To match the required form, write $1 + (k-1)\lambda = (1 + k\lambda) - \lambda$:

$$\begin{aligned}\left(1 + (k-1)\lambda\right)\left(1 - \mathrm{e}^{-\lambda}\right) - \lambda \mathrm{e}^{-\lambda} &= (1 + k\lambda)\left(1 - \mathrm{e}^{-\lambda}\right) - \lambda + \lambda \mathrm{e}^{-\lambda} - \lambda \mathrm{e}^{-\lambda} \\ &= (1 + k\lambda)\left(1 - \mathrm{e}^{-\lambda}\right) - \lambda.\end{aligned}$$

With $\lambda = \mu p$ this gives

$$Q = \mathrm{e}^{-\mu p(k-1)}\left\{(1 + k\mu p)\left(1 - \mathrm{e}^{-\mu p}\right) - \mu p\right\},$$

which is the stated result.

Checks

- For $k = 1$ the formula gives $1 - \mathrm{e}^{-\lambda} - \lambda \mathrm{e}^{-\lambda} = \mathrm{P}(M_1 \geq 2)$, as it must.
- The probabilities should sum to 1 over $k = 1, 2, \ldots$ (she marks a second cell eventually with probability 1); summing numerically for several values of λ confirms this.
- A simulation with $\mu = 1.6$, $p = 0.45$ (so $\lambda = 0.72$) gave the second mark on slide 2 in about 26.0% of 200 000 runs; the formula gives $Q \approx 0.2591$.

Common slips

- Forgetting the case "one mark before slide k, at least one on slide k", or including $S = 1$, $M_k = 0$ by mistake.
- Working slide by slide with "first slide with a mark" arguments, which gets messy; the total S over the first $k - 1$ slides is the efficient object.
- Not saying where independence (between slides, and between S and M_k) is used.

Beyond STEP

Thinning is one face of the Poisson process: if events occur as a Poisson process and each is kept independently with probability p, the kept events again form a Poisson process, with rate scaled by p. Part (ii) is then a discrete version of the waiting time for the second event, which in continuous time has a gamma distribution. Grimmett and Welsh, *Probability: An Introduction*, develops Poisson processes and these waiting-time distributions.

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